Noncommutative Geometric Structure of Subhomogeneous Operator Systems

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Outline

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Compact matrix convex sets

Definition (Matrix convex set)

Let V be a locally compact topological vector space. A collection $K = \{K_n\}_{n=1}^{\infty}$ such that $K_n \subset \mathbb{M}_n(V)$ is **matrix convex** if

$$\sum_i \gamma_i^* v_i \gamma_i \in K_n,$$

for all $v_i \in K_{n_i}$ and $\gamma_i \in \mathbb{M}_{n_i,n}$ satisfying $\sum_i \gamma_i^* \gamma_i = I_n$.

- The sum $\sum_i \gamma_i^* v_i \gamma_i$ is called a **matrix convex combination** in K over V.
- When γ_i is surjective for all i, we say the combination is **proper**.
- If K_n is compact for all $n \in \mathbb{N}$, we say K is **compact**.
- If K, W are matrix convex sets, a collection $F = \{F_n\}_{n=1}^{\infty}$ of mappings $F_n : K_n \to W_n$ is **matrix affine** if

$$F_n\left(\sum_i \gamma_i^* v_i \gamma_i\right) = \sum_i \gamma_i^* F_{n_i}(v_i)\gamma_i,$$

for all $v_i \in K_{n_i}$ and $\gamma_i \in \mathbb{M}_{n_i,n}$ satisfying $\sum_i \gamma_i^* \gamma_i = I_n$.

Compact matrix convex sets

- A matrix affine map $F : K \to W$ is **continuous** if F_n is continuous for all $n \in \mathbb{N}$.
- If F is continuous, invertible and has a continuous inverse, we say F is a **matrix affine homeomorphism**.

Definition (Matrix extreme points)

Let K be a compact matrix convex set. An element $v \in K_n$ is a **matrix extreme point** if whenever $v = \sum_i \gamma_i^* v_i \gamma_i$ is a proper matrix convex combination of $v_i \in K_{n_i}$, then for all i, $n_i = n$ and v_i is unitarily equivalent to v.
v is **absolute matrix extreme** if whenever $v = \sum_i \gamma_i^* v_i \gamma_i$ is a matrix convex combination of $v_i \in K_{n_i}$ with γ_i non-zero, then for all i,

- if $n_i \leq n$, then $n_i = n$ and v_i is unitarily equivalent to v.
- if $n_i > n$, then v_i is unitarily equivalent to $v \oplus \rho_i$ with $\rho_i \in K_{n_i - n}$.

- The set of matrix extreme points of K is denoted by ∂K.
- The set of absolute matrix extreme points of K is denoted by $\mathrm{Abex}(K)$.

Compact matrix convex sets

- There are many examples for sets K with $\mathrm{Abex}(K) = \emptyset$.

Theorem (Webster-Winkler, '99)

Let K be a compact matrix convex set. Then,

$$K = \overline{\mathrm{matconv}}(\partial K)\,.$$

- K is called **n-subhomogeneous** if n is the smallest integer for which $\partial K \subset \bigcup_{i=1}^{n} K_n$.
- n-subhomogeneous sets are generated by K_n,

$$K = \overline{\mathrm{matconv}}(K_n)\,.$$

Operator systems

- Let $\mathcal{A}$ be a C^*-algebra. An **operator system** is a unital subspace $\mathcal{S} \subset \mathcal{A}$ which is closed under the adjoint operation.
- A linear map $\phi : \mathcal{S} \to \mathcal{R}$ between two operator systems is called **n-positive** if

$$\forall\,(s_{ij}) \in \mathbb{M}_n(\mathcal{S}), \quad (s_{ij}) \geq 0 \Longrightarrow (\phi(s_{ij})) \geq 0.$$

- ϕ is **completely positive** if ϕ is n-positive for all $n \in \mathbb{N}$.
- $\mathrm{UCP}(\mathcal{S}, \mathcal{R})$ denotes the set of all unital and completely positive maps from $\mathcal{S}$ to $\mathcal{R}$.

Definition (Matrix state space)

Let $\mathcal{S}$ be an operator system. For all $n \in \mathbb{N}$, Let $\mathrm{UCP}_n(\mathcal{S}) := \mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$. The **matrix state space** of $\mathcal{S}$ is the set

$$\mathrm{UCP}(\mathcal{S}) := \bigcup_{n \in \mathbb{N}} \mathrm{UCP}_n(\mathcal{S}).$$

- $\mathrm{UCP}(\mathcal{S})$ is compact matrix convex set over $\mathcal{S}^*$.

Matrix state space and the matrix range

Definition (Matrix range)

Let $S = (S_1, \ldots, S_d) \in \mathrm{B}(\mathcal{H})^d$ and let $\mathcal{S}(S) = \mathrm{span}(S, S^*, I)$. The **matrix range** of S is the set

$$\mathcal{W}(S) = \bigcup_{n=1}^{\infty} \mathcal{W}^n(S), \quad \mathcal{W}^n(S) := \{(\phi(S_1), \ldots, \phi(S_d)) | \phi \in \mathrm{UCP}_n(\mathcal{S}(S))\} \subset \mathbb{M}_n^d.$$

- $\mathcal{W}(S)$ is a compact matrix convex set over $\mathbb{C}^d$.
- If $\mathcal{S} = \mathcal{S}(S)$, then $\mathrm{UCP}(\mathcal{S})$ is matrix affine homeomorphic to $\mathcal{W}(S)$.
- [K. '23] For finitely generated subhomogeneous operator systems of order at most n, $\mathcal{W}^n(S) = \mathcal{W}^n(T)$ implies $\mathcal{W}(S) = \mathcal{W}(T)$.

C^*-envelopes

Definition (Boundary representation)

Let $\mathcal{S}$ be an operator system. A $*$-representation $\pi : \mathrm{C}^*(\mathcal{S}) \to \mathrm{B}(\mathcal{H})$ is a **boundary representation** if π is irreducible and π is the unique UCP extension of $\pi|_{\mathcal{S}}$.

Definition (C^*-cover and C^*-envelope)

Let $\mathcal{S}$ be an operator system. A **C^*-cover** of $\mathcal{S}$ is a pair $(\mathcal{A}, \iota)$ of a unital C^*-algebra $\mathcal{A}$ and a unital completely isometric map $\iota : \mathcal{A} \to \mathrm{C}^*(\iota(\mathcal{S}))$.
A **C^*-envelope** of $\mathcal{S}$ is a C^*-cover $(\mathrm{C}^*_{\mathrm{e}}(\mathcal{S}), \iota)$ of $\mathcal{S}$ which is minimal in the sense that if $(\mathcal{A}, \psi)$ is a C^*-cover of $\mathcal{S}$, there is a unique surjective $*$-homomorphism $\pi : \mathcal{A} \to \mathrm{C}^*_{\mathrm{e}}(\mathcal{S})$ such that $\pi \circ \psi = \iota$ for all $s \in \mathcal{S}$.

- [Hamana, '07] Every operator system has a C^*-envelope.
- The C^*-envelope is unique up to isomorphism.
- [Davidson-Kennedy, '15] If $\rho := \bigoplus_{[\pi] \in \mathrm{bdry}(\mathcal{S})} \pi$ and $\mathcal{A} = \mathrm{C}^*(\rho(\mathcal{S}))$, then $\mathcal{A} = \mathrm{C}^*_{\mathrm{e}}(\mathcal{S})$.

Matrix extreme points in $\mathrm{UCP}(\mathcal{S})$

Definition (Subhomogeneous operator system)

An operator system $\mathcal{S}$ is **n-subhomogeneous** if $\mathrm{UCP}(\mathcal{S})$ is n-subhomogeneous.

- [Farenick, '00] The matrix extreme points of $\mathrm{UCP}(\mathcal{S})$ are precisely the pure maps.
- [Arveson, '07] The absolute matrix extreme points of $\mathrm{UCP}(\mathcal{S})$ are pure and maximal maps. In particular, absolute matrix extreme points extend to boundary representation of $\mathrm{C}^*(\mathcal{S})$ for $\mathcal{S}$.
- [Davidson-Kennedy, '15] A matrix extreme point $\phi \in \partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$ is either absolute, or it dilates to another matrix extreme point $u \in \partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_{n+1})$.
- [Davidson-Kennedy, '15] A matrix extreme point $\phi \in \partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$ dilates to a restriction of a boundary representation on a (possibly infinite dimensional) Hilbert space.

The C^*-envelope of a subhomogeneous operator system

- Let $\mathcal{S}$ be an operator system for which $C^*(\mathcal{S})$ is an n-subhomogeneous C^*-algebra.
- Every boundary representation of $C^*(\mathcal{S})$ for $\mathcal{S}$ is of dimension at most n.
- Every matrix extreme point dilates to a maximal matrix extreme point, which is in level at most n.
- Therefore, $\mathcal{S}$ is n-subhomogeneous.
- The converse does not hold in general.

Example (The Toeplitz algebra)

Let $\mathcal{T}$ be the C^*-algebra generated by the unilateral shift S on $\ell^2(\mathbb{N})$. Then,

- $\mathcal{T}$ is irreducible and infinite dimensional, so it is not subhomogeneous.
- Boundary representations of $\mathcal{T}$ are all 1 dimensional, so $\mathcal{S}(S)$ is 1-subhomogeneous.

The C^*-envelope of a subhomogeneous operator system

Theorem (K, '25)

Let $\mathcal{S}$ be an n-subhomogeneous operator system. Then, $\mathrm{C}^*_{\mathrm{e}}(\mathcal{S})$ is n-subhomogeneous.

Sketch of proof.

Define

$$\rho = \bigoplus_{\pi \in \mathrm{bdry}_{\leq \mathrm{n}}(\mathcal{S})} \pi.$$

- $(\mathrm{C}^*(\rho(\mathcal{S})), \rho)$ is a C^*-cover for $\mathcal{S}$.
- $\mathrm{C}^*(\rho(\mathcal{S}))$ is n-subhomogeneous.
- Let $\rho : \mathrm{C}^*(\rho(\mathcal{S})) \to \mathrm{C}^*_{\mathrm{e}}(\mathcal{S})$ be the surjective homomorphism. If $\pi \in \mathrm{irrep}(\mathrm{C}^*_{\mathrm{e}}(\mathcal{S}))$, then $\pi \circ \rho \in \mathrm{irrep}(\mathrm{C}^*(\rho(\mathcal{S})))$ and thus of dimension at most n.

□

The constrained d-shift

- Let $\mathcal{H}_2^d$ be the completion of $\mathbb{C}[z_1, \ldots, z_d]$ with respect to the inner product for which

$$\left\langle z^\alpha, z^\beta \right\rangle = \delta_{\alpha,\beta} \frac{\alpha!}{|\alpha|!}.$$

- If $I \lhd \mathbb{C}[z_1, \ldots, z_d]$ is a homogeneous ideal, we set $\mathcal{F}_I = \mathcal{H}_2^d \ominus I$ and

$$S = (S_1, \ldots, S_d), \quad S_i = P_{\mathcal{F}_I} M_{z_i}|_{\mathcal{F}_I},$$

  where M_{z_i} is the multiplication operator by z_i and $P_{\mathcal{F}_I}$ is the orthogonal projection onto $\mathcal{F}_I$.
- We also set

$$\mathcal{S}_I := \operatorname{span}(S, S^*, I), \quad \mathcal{A}_I := \overline{\operatorname{alg}}(S), \quad \mathcal{T}_I := \mathrm{C}^*(S).$$

- We call the d-tuple S the **constrained d-shift** with respect to I.

The constrained d-shift

- S is a row contraction - $\sum_i S_i^* S_i \leq 1$.
- S satisfies the polynomial relations $p(S) = 0$ for all $p \in I$.
- [Popescu, 2006] For any row contraction T such that $p(T) = 0$ for all $p \in I$, there exists an completely contractive homomorphism

$$\pi : \mathcal{A}_I \to \overline{\mathrm{alg}}(T), \quad \pi(S_i) = T_i,\ i = 1, \ldots, d.$$

Proposition (K, '25)

Let $\mathcal{W}(S)$ be the matrix range of the constrained d-shift. Then, $\mathcal{W}(S)$ is not subhomogeneous.

The constrained d-shift

Proof.

[Davidson, Ramsey, Shalit, '11] $\mathcal{T}_I = \mathrm{C}^*_{\mathrm{e}}(S)$ and the identity map is irreducible. Therefore, $\mathcal{S}_I$ is an operator system which has a non-subhomogeneous C^*-envelope. Therefore, $\mathcal{S}_I$ is not subhomogeneous, and $\mathcal{W}(S)$ is not subhomogeneous. □

- Although $\mathcal{W}(S)$ is not subhomogeneous, we can still study its geometric structure by considering its matrix extreme structure. Consider the set $\overline{V}(I)$ where

$$V(I) := \left\{ X \in \mathbb{M}_n^d \,\middle|\, \substack{X \text{ is a row contraction} \\ p(X)=0,\, \forall p \in I} \right\}.$$

- For all $X \in \overline{V}(I)$, the mapping $S \mapsto X$ extends to a completely positive map of $\mathcal{S}_I$ into $\mathcal{S}(X)$, thus $\overline{V}(I) \subset \mathcal{W}(S)$.

The constrained d-shift

Theorem (K, '25)

$\partial\mathcal{W}(S) \subset \overline{\mathrm{matconv}}(V(I))$. *In particular,* $\mathcal{W}(S) = \overline{\mathrm{matconv}}(V(I))$.

Sketch of Proof

[Davidson-Kennedy, '15] An extreme point X of $\mathcal{W}(S)$ is a compression of $\pi(S)$ where π is a boundary representation of $\mathcal{T}_I$ for $\mathcal{S}_I$. If π is finite dimensionl,

$$X = V^*\pi(S)V \in \mathrm{matconv}(V(I))\,.$$

If π is infinite dimensional, set

$$S_n = (S_{n,1}, \ldots, S_{n,d})\,, \quad S_{n,i} = P_{\mathcal{N}_n}\, S_i|_{\mathcal{N}_n}\,,$$

and $S_n \in V(I)$ for all $n \in \mathbb{N}$ such that $S_n \xrightarrow{SOT} S$.

The constrained d-shift

Sketch of Proof.

Then,

$$X = \mathrm{SOT}\text{-}\lim_{n\to\infty} V^* S_n V \in \overline{\mathrm{matconv}}(V(I))\,.$$

From this it follows that $\partial\mathcal{W}(S) \subset \overline{\mathrm{matconv}}(V(I))$.By [Webster-Winkler,'99]

$$\mathcal{W}(S) = \overline{\mathrm{matconv}}(\partial\mathcal{W}(S)) \subset \overline{\mathrm{matconv}}(V(I)) \subset \mathcal{W}(S),$$

which completes the proof. □

Thank You!

Questions?

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